

Chapter 3

Demand, Supply and Price

Chapter Outline/Learning Objectives

Section	Learning Objectives
	After studying this chapter, you will be able to
3.1 Demand	1. list the factors that determine the quantity demanded of a good. 2. distinguish between a shift of the demand curve and a movement along the demand curve.
3.2 Supply	3. list the factors that determine the quantity supplied of a good. 4. distinguish between a shift of the supply curve and a movement along the supply curve.
3.3 The Determination of Price	5. explain the forces that drive market price to equilibrium, and how equilibrium price is affected by changes in demand and supply.

3.1 Demand

- **Quantity Demanded**
 - The total amount that consumers desire to purchase in some time period is called the **quantity demanded** of a product.
 - Quantity bought (or exchanged) refers to **actual** purchases.
 - Quantity demanded is a **flow**, as opposed to a **stock**.

Quantity Demanded and Price

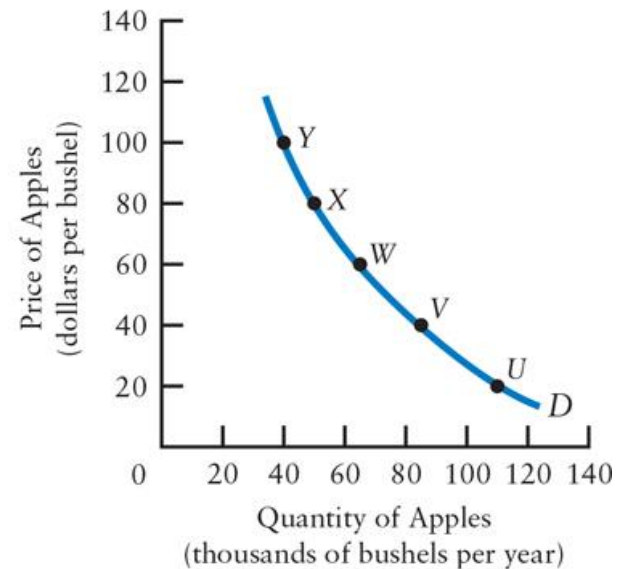
- A basic hypothesis is that – *ceteris paribus* – the price of a product and the quantity demanded are **negatively** related.
- Why? There are usually several products that can satisfy any given want or desire.
- A reduction in the price of a product means that the specific desire can now be satisfied more cheaply by buying more of that product.

Demand Schedules and Demand Curves (1 of 3)

A Demand Schedule for Apples

Reference Point	Price (\$ per bushel)	Quantity Demanded (thousands of bushels per year)
U	20	110
V	40	85
W	60	65
X	80	50
Y	100	40

A Demand Curve for Apples



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Figure 3-1 The Demand for Apples

Demand Schedules and Demand Curves (2 of 3)

- A change in variables **other than price** will shift the demand curve to a new position.
 - Consumer's income
 - Prices of other goods
 - Consumers' preferences
 - Population
 - Significant changes in weather

Demand Schedules and Demand Curves (3 of 3)

Demand Schedules

Price (\$ per bushel)	Quantity Demanded (thousands of bushels per year when average annual income is \$50 000)	Quantity Demanded (thousands of bushels per year when average annual income is \$60 000)
p	D_0	D_1
20	110 U	140 U'
40	85 V	115 V'
60	65 W	95 W'
80	50 X	80 X'
100	40 Y	70 Y'

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Demand Curves

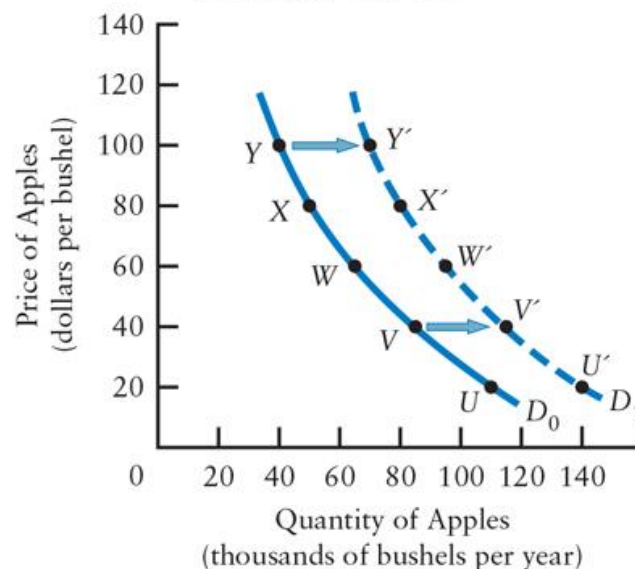


Figure 3-2 An Increase in the Demand for Apples

Figure 3-3 Shifts in the Demand Curve

- A **rightward** shift indicates an **increase** in demand.
- A **leftward** shift indicates a **decrease** in demand.

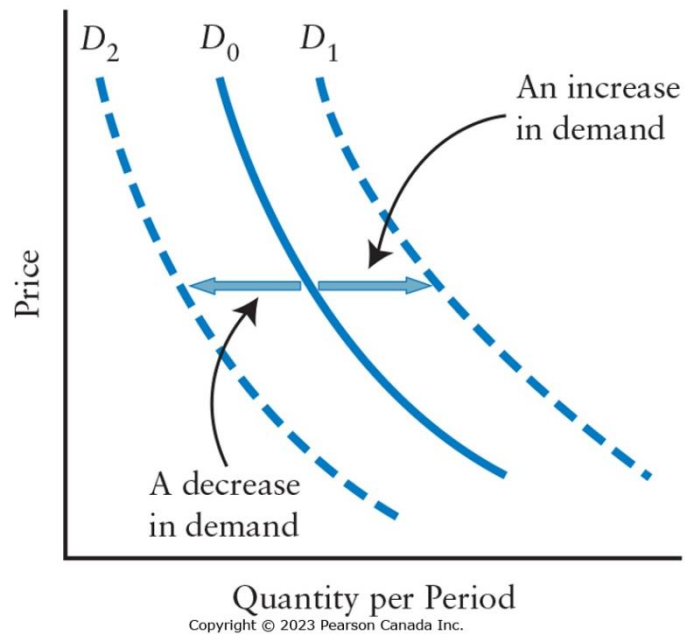
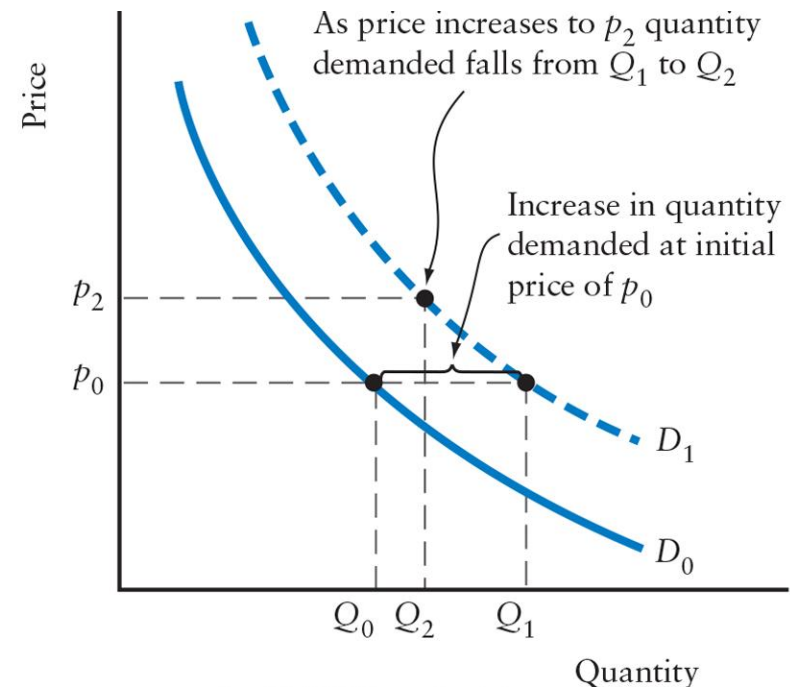


Figure 3-4 Shifts of and Movements Along the Demand Curve

- A **change in demand** is a change in quantity demanded at every price – a shift of the entire curve.
- A **change in quantity demanded** refers to a movement from one point on a demand curve to another point – a movement along the demand curve.



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3.2 Supply

- **Quantity Supplied**

- The amount of a product that firms **desire** to sell in some time period is called the **quantity supplied** of that product.
- Quantity supplied is the amount that firms are willing to offer for sale and not necessarily the **quantity actually sold**.
- Quantity supplied is a flow as opposed to a stock.

Quantity Supplied and Price

- A basic hypothesis is that—*ceteris paribus*—the price of the product and the quantity supplied are **positively** related.
- Why? Producers are interested in making profits.
- If the price of a particular product rises, then the production and sale of this product is more profitable.

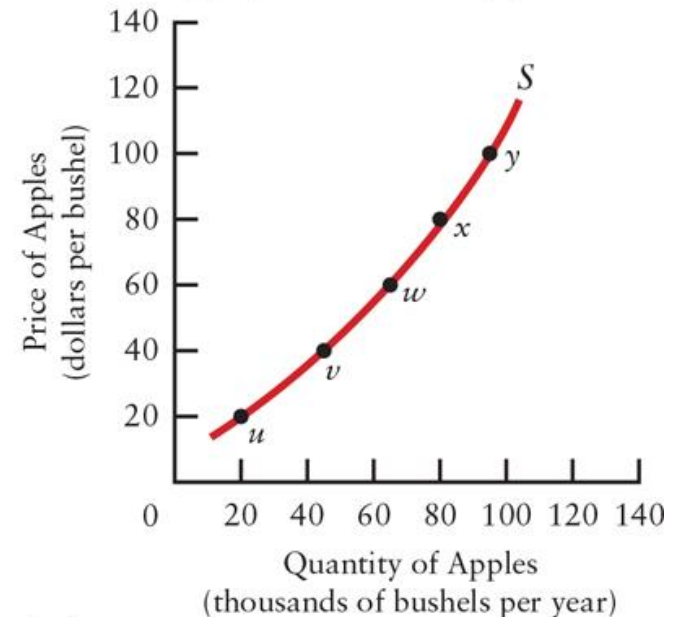
Supply Schedules and Supply Curves

(1 of 3)

A Supply Schedule for Apples

Reference Point	Price (\$ per bushel)	Quantity Supplied (thousands of bushels per year)
<i>u</i>	20	20
<i>v</i>	40	45
<i>w</i>	60	65
<i>x</i>	80	80
<i>y</i>	100	95

A Supply Curve for Apples



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Figure 3-5 The Supply of Apples

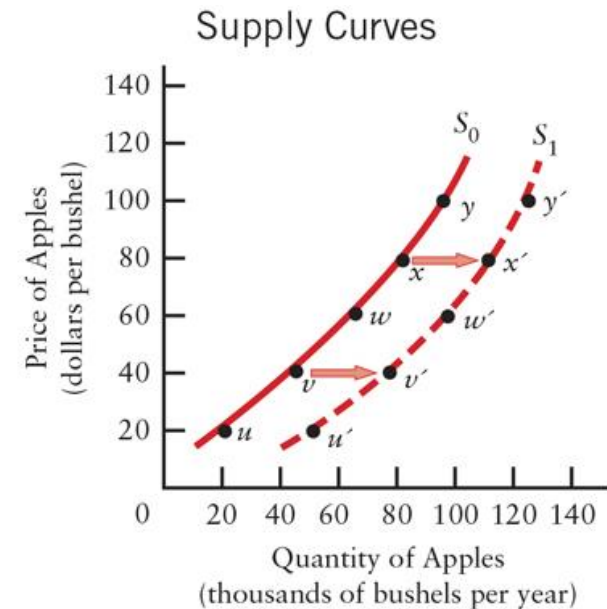
Supply Schedules and Supply Curves

(2 of 3)

- A change in any variable **other than price** will shift the supply curve to a new position.
 - Prices of inputs
 - Technology
 - Government taxes or subsidies
 - Prices of other products
 - Significant changes in weather
 - Number of suppliers

Supply Schedule and Supply Curve (3-3)

Supply Schedules		
Price (\$ per bushel)	Quantity Supplied (thousands of bushels per year before innovation)	Quantity Supplied (thousands of bushels per year after innovation)
p	S_0	S_1
20	20 u	50 u'
40	45 v	75 v'
60	65 w	95 w'
80	80 x	110 x'
100	95 y	125 y'



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Figure 3-6 An Increase in the Supply of Apples

Shifts of and Movements Along the Supply Curve

- A **change in supply** is a change in quantity supplied at every price – a shift of the entire curve.
- A **change in quantity supplied** refers to a movement from one point on a supply curve to another point – a movement along the supply curve.

Applying Economic Concepts 3-1

- **Demand and Supply Shocks Created by the COVID-19 Pandemic**
 - Due to lockdowns the demand for certain goods and services declined, while it increased in other areas.
 - Various restrictions also affected supply of goods and services in some industries.
 - To provide income relief for the millions of Canadians whose regular incomes had suddenly disappeared, the government increased its spending financed by borrowing.
 - When governments borrow, they issue new bonds (IOUs).

3.3 The Determination of Price

- **The Concept of a Market**

- A market may be defined as any situation in which buyers and sellers negotiate the transaction of some goods or services.
- Markets may differ in the degree of **competition** among various buyers and sellers.
- In a **perfectly competitive market** buyers and sellers are **price takers**.

Market Equilibrium

- At the **equilibrium price** (\$60), quantity demanded is equal to quantity supplied (65,000 bushels).
- At any price above \$60, there is excess supply and thus downward pressure on price.
- At any price below \$60, there is excess demand and thus upward pressure on price.
- Market “clears” at equilibrium.

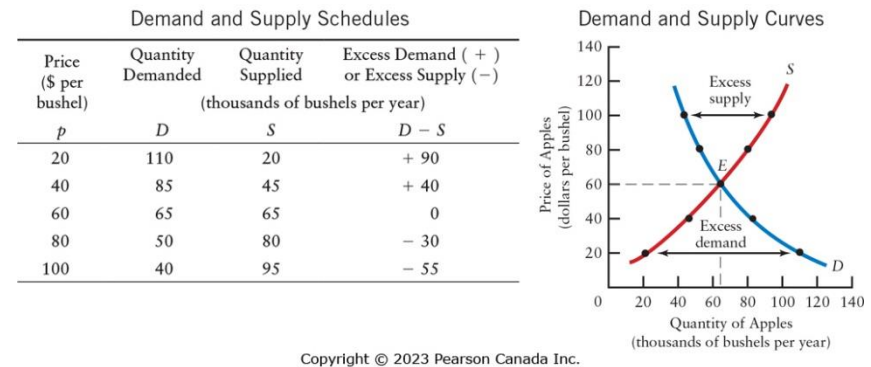
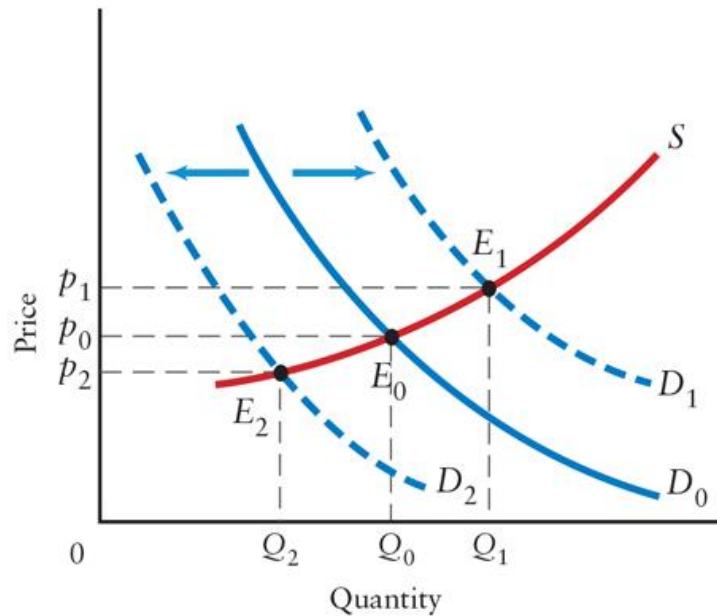


Figure 3-7 The Equilibrium Price of Apples

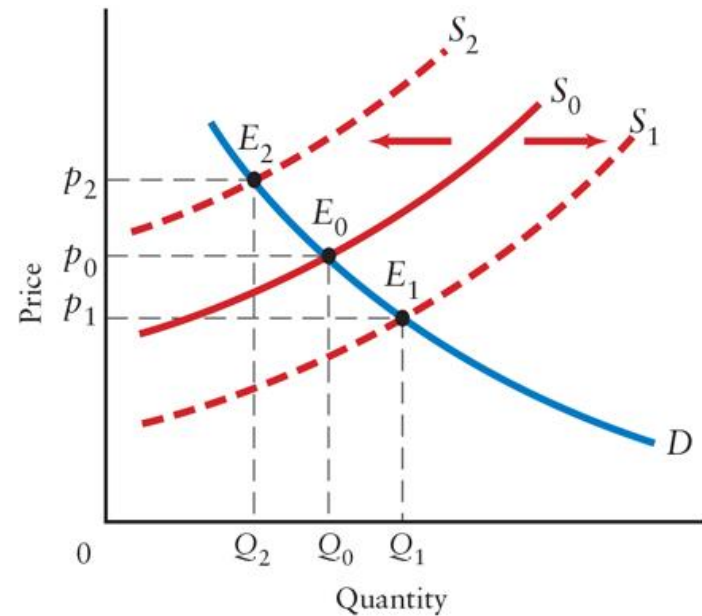
Changes in Market Equilibrium

- The four possible curve shifts:
 - **An increase in demand** causes an increase in both the equilibrium price and equilibrium quantity.
 - **A decrease in demand** causes a decrease in both equilibrium price and equilibrium quantity.
 - **An increase in supply** causes a decrease in the equilibrium price and an increase in the equilibrium quantity.
 - **A decrease in supply** causes an increase in the equilibrium price and a decrease in the equilibrium quantity.

Changes in Market Equilibrium (2 of 2)



(i) The effect of shifts in the demand curve



(ii) The effect of shifts in the supply curve

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Figure 3-8 Shifts in Demand and Supply Curves

Relative Prices and Inflation

- The **absolute price** of a product is the amount of money that must be spent to acquire one unit of that product.
- A **relative price** is the price of one good in terms of another.
- Demand and supply curves are drawn in terms of **relative prices** rather than **absolute prices**.

Word of Caution

- The theory of demand and supply can be used to explain changes in supply and demand in many markets.
- But the model has important limitations and cannot be usefully applied to the markets for a number of consumer products.
- To understand why, see *Applying Economics Concepts 3-2*.

Applying Economic Concepts 3-2

- **Why Apples but Not iPhones?**
- Three conditions must be satisfied for price determination in a market to be well described by the demand-and-supply model:
 1. Large number of consumers; each one small relative to the size of the market.
 2. Large number of producers; each one small relative to the size of the market.
 3. Producers must be selling 'homogeneous' versions of the product.